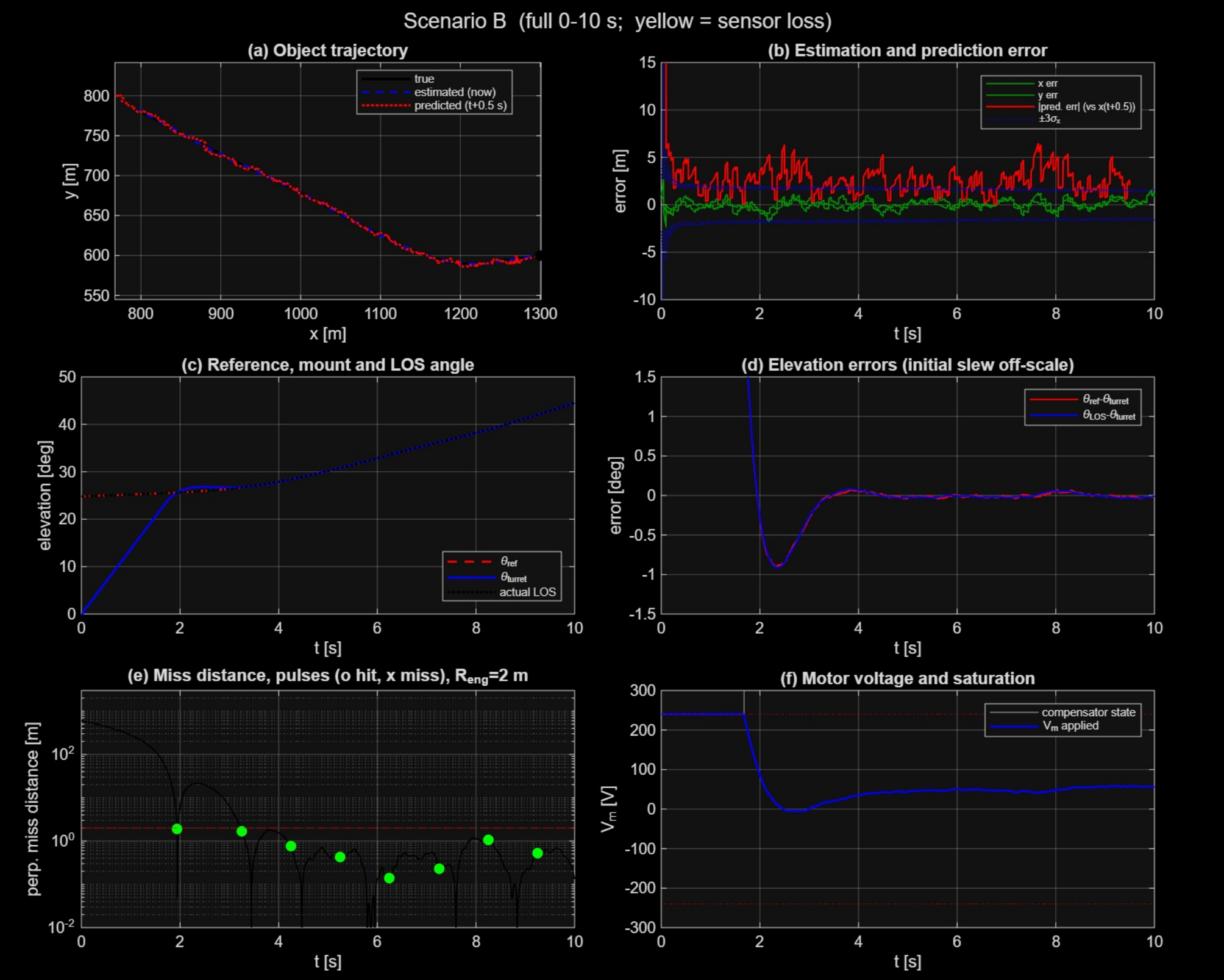

Scenario B (full 0-10 s; yellow = sensor loss)
(a) Object trajectory
true
estimated (now)
predicted (t+0.5 s)
y [m]
x [m]
(b) Estimation and prediction error
x err
y err
|pred. err| (vs x(t+0.5))
±3σx
error [m]
t [s]
(c) Reference, mount and LOS angle
θref
θturret
actual LOS
elevation [deg]
t [s]
(d) Elevation errors (initial slew off-scale)
θref-θturret
θLOS-θturret
error [deg]
t [s]
(e) Miss distance, pulses (o hit, x miss), Reng=2 m
perp. miss distance [m]
t [s]
(f) Motor voltage and saturation
compensator state
Vm applied
Vm [V]
t [s]